An Exhaustive Named-Coordinate Atlas for a Universal Complement Core

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Abstract

The common intersection of all complements to a fixed direct factor in an exact restricted mark cokernel was previously identified as $8C \cong \mathbb{Z}/3 \oplus \mathbb{Z}/18$. We resolve the full incidence of the sixteen named presentation classes inside this core. An explicit $\mathbb{Z}/9 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/2$ coordinate model classifies all $2^{16} = 65536$ named supports. They reach exactly twenty distinct subgroups, and these are every subgroup of the core. We give the complete ten-type distribution by support size and the generating-support polynomial. There are exactly twenty-five inclusion-minimal generating supports, all triples containing the ninth named class. This is a presentation-dependent atlas, not a canonical coordinate system.

1 Core, scope, and coordinate model

Let M be the frozen 16×16 restricted mark matrix and write $[e_j]$ for its named coordinate classes. The predecessor theorem identifies the universal complement core as

$$Q = 8C \cong \mathbb{Z}/3 \oplus \mathbb{Z}/18, \qquad |Q| = 54.$$

We study the labelled family $q_j = 8[e_j]$ for $1 \le j \le 16$.

The scope firewall is

`NO_BAD_EULER_OR_ROOT_NUMBER`.

No canonical Smith coordinates, full table of marks or Burnside ring, arithmetic or local data, Euler factor, root number, automorphy, or Hilbert–Polya operator is claimed.

Exact rational residues modulo $M\mathbb{Z}^{16}$ show that (q_1, q_3, q_9) gives coordinates on

$$Q = \mathbb{Z}/9 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/2.$$

In this basis the sixteen rows are

$$\begin{aligned} &(1,0,0), (6,0,0), (0,1,0), (3,1,0), (0,0,0), (0,0,0),\\ &(4,2,0), (3,2,0), (0,0,1), (0,0,0), (0,1,0), (3,1,0),\\ &(0,0,0), (0,0,0), (2,1,0), (8,2,0). \end{aligned} \tag{1}$$

For each row, the checker lifts the asserted relation back to an integral column of M. Thus Equation (1) is tied to the original presentation, not inferred only from element orders.

2 The complete subgroup atlas

For $A \subseteq \{1, \dots, 16\}$ put

$$Q_A = \langle q_j : j \in A \rangle.$$

The producer closes all 65536 supports in the coordinate model. A separate queue enumeration starts from zero, adjoins every element of the 54-element group, and returns its complete subgroup lattice.

Theorem 1 (Exact lattice coverage)**.** *The family $\{Q_A : A \subseteq \{1, \dots, 16\}\}$ consists of twenty distinct subgroups and equals the complete subgroup lattice of Q. The type inventory is*

type	1	C_2	C_3	C_6	C_3^2	C_9	C_{18}	$C_3{+}C_6$	$C_3{+}C_9$	$C_3{+}C_{18}$
#	1	1	4	4	1	3	3	1	1	1.

Proof. Cached subgroup closure applied to Equation (1) produces twenty actual subsets of Q. The independent all-element enumeration also produces twenty subsets, and exact set comparison gives equality. GAP independently enumerates the same ten isomorphism types with the displayed multiplicities. $\square$

Table 1 refines the result by support size. Its columns use the type order in the theorem. Thus it records every one of the 65536 supports, while keeping supports that generate the same subgroup distinct.

$\lvert A\rvert$	1	C_2	C_3	C_6	C_3^2	C_9	C_{18}	$C_3{+}C_6$	$C_3{+}C_9$	Q	total
0	1	0	0	0	0	0	0	0	0	0	1
1	5	1	6	0	0	4	0	0	0	0	16
2	10	5	32	6	13	25	4	0	25	0	120
3	10	10	70	32	85	66	25	13	224	25	560
4	5	10	80	70	245	95	66	85	940	224	1820
5	1	5	50	80	411	80	95	245	2461	940	4368
6	0	1	16	50	446	39	80	411	4504	2461	8008
7	0	0	2	16	328	10	39	446	6095	4504	11440
8	0	0	0	2	165	1	10	328	6269	6095	12870
9	0	0	0	0	55	0	1	165	4950	6269	11440
10	0	0	0	0	11	0	0	55	2992	4950	8008
11	0	0	0	0	1	0	0	11	1364	2992	4368
12	0	0	0	0	0	0	0	1	455	1364	1820
13	0	0	0	0	0	0	0	0	105	455	560
14	0	0	0	0	0	0	0	0	15	105	120
15	0	0	0	0	0	0	0	0	1	15	16
16	0	0	0	0	0	0	0	0	0	1	1

Table 1: Complete named-support distribution. Each row sums to $\binom{16}{|A|}$. The penultimate type is $C_3 \oplus C_9$ and the last type column Q is $C_3 \oplus C_{18}$.

3 The generating-support complex

Reading the Q column of Table 1 gives the polynomial

$$\begin{aligned} P_Q(t) = 25t^3 + 224t^4 + 940t^5 + 2461t^6 + 4504t^7 + 6095t^8 \\ + 6269t^9 + 4950t^{10} + 2992t^{11} + 1364t^{12} \\ + 455t^{13} + 105t^{14} + 15t^{15} + t^{16}. \end{aligned} \tag{2}$$

Proposition 1 (Minimal named supports)**.** *Exactly twenty-five named supports are inclusion-minimal among those that generate Q. Every one has size three and contains S_9.*

Proof. The $\mathbb{Z}/2$ coordinate in Equation (1) is nonzero only in row nine, so every generating support contains S_9. After choosing it, generation is equivalent to spanning the Frattini quotient of $\mathbb{Z}/9 \oplus \mathbb{Z}/3$ over $\mathbb{F}_3$. Exactly twenty-five unordered pairs of the remaining named rows have nonzero determinant. Removing any member then loses either the dyadic coordinate or the required rank, proving inclusion-minimality. Exhaustive deletion independently returns the same twenty-five triples. $\square$

The number three is a property of the named list: no one or two of the q_j generate Q. It is not the abstract generator rank. Indeed $Q \cong \mathbb{Z}/3 \oplus \mathbb{Z}/18$ is abstractly two-generated. Likewise, the coordinate basis is a verified choice, not a canonical Smith basis.

4 Reproducibility and conclusion

The canonical evidence binds the exact C64 matrix and C71 evidence and manifest. Producer and checker independently classify all supports and all abstract subgroups; the checker also verifies every coordinate relation in the original integer presentation. GAP reproduces the complete subgroup type inventory. Clean replay passes and twenty-eight hostile semantic mutations are rejected. C72 thus upgrades the predecessor's list of twenty-five triples to a complete labelled incidence atlas: every named support is classified, and every subgroup of the universal core is reached.